

MDI3003

ADVANCED PREDICTIVE ANALYTICS

LAB-06

**CRIME INCIDENT FORECASTING USING AUTOREGRESSIVE AND ARIMA
MODELS**

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COURSE TITLE : Advanced Predictive Analytics

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GithubLink: https://github.com/Harshita2011/Advanced_Predictive_Analytics_LAB

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1.Aim

To analyze the temporal patterns of reported crime incidents for a selected location and develop Autoregressive (AR) and ARIMA-based time-series forecasting models to predict future incident counts using a chronological, leakage-safe evaluation framework.

2.Objectives

The objectives of this experiment are to:

1. To construct a regular time-series of reported incident counts by aggregating incident-level records for the selected location at an appropriate time frequency.
2. To perform exploratory time-series analysis to identify trends, seasonal patterns, variability, autocorrelation, outliers, and other temporal characteristics.
3. To divide the time series into chronological training and test periods while preventing temporal data leakage and avoiding random shuffling.
4. To assess the stationarity of the training series using visual analysis and the Augmented Dickey-Fuller (ADF) test.
5. To examine ACF and PACF diagnostics to understand temporal dependence and support the selection of appropriate model parameters.
6. To establish a Naive persistence baseline and compare its forecasting performance with AR and ARIMA models.
7. To develop and evaluate AR and ARIMA forecasting models, with model orders selected using information available only from the training/validation data.
8. To evaluate the forecasting performance of the models on the locked future test period using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).
9. To analyze model residuals using residual plots, residual autocorrelation, and the Ljung-Box test to assess the adequacy of the fitted models.
10. To assess the robustness and stability of the forecasting approach using rolling-origin/walk-forward validation and replication across an additional location, where applicable.
11. To interpret the forecasting results while recognizing that the models estimate reported/recorded incident counts and do not represent actual crime prevalence or individual-level criminal behavior.
12. To document the data-processing, modeling, evaluation, and experimental settings required to ensure reproducibility of the analysis and results.

3. Executive Summary

This report presents a comprehensive evaluation of three short-horizon time-series forecasting approaches—Naive Persistence, Autoregressive (AR), and ARIMA models—for forecasting reported crime-incident counts. The analysis was conducted using two independently executed public-safety time series: the Chicago Crime Series, representing Police District 12 with daily aggregation over 251 days, and the Brooklyn Complaint Series, aggregated weekly over 79 weeks. The use of two datasets with different locations, reporting agencies, aggregation frequencies, and temporal scales provides a broader evaluation of the applicability and consistency of the forecasting methodology.

A consistent and chronological, leakage-safe forecasting framework was implemented across both datasets. The methodology included systematic time-series construction, exploratory analysis, stationarity assessment, ACF/PACF diagnostics, model development, training-based model selection, rolling-origin validation, and final evaluation on a locked future test period. A Naive persistence model was established as a benchmark, followed by AR and ARIMA models, allowing the forecasting capability of each approach to be assessed under a common experimental framework.

The forecasting results demonstrate that both simple and advanced statistical approaches can provide useful short-horizon forecasts depending on the characteristics of the underlying series. For the Chicago series, the Naive model achieved the strongest forecasting performance, obtaining an MAE of 7.07 incidents/day and an RMSE of 11.00 incidents/day. For the Brooklyn series, ARIMA(2,1,2) achieved the best performance, with an MAE of 354.06 incidents/week and an RMSE of 830.27 incidents/week. These results demonstrate the effectiveness of persistence-based forecasting as well as the ability of ARIMA to capture additional temporal structure where appropriate.

The methodology was further validated through replication across ten geographic locations, comprising multiple Chicago districts and New York City boroughs. The results showed that the Naive model achieved the lowest forecasting error in six locations, while ARIMA achieved the lowest error in the remaining four locations. This provides encouraging evidence that the forecasting framework can be applied across multiple geographic units while maintaining a consistent evaluation methodology.

An important strength of the analysis is its emphasis on reproducibility and systematic evaluation. Dataset selection, filtering, aggregation frequency, chronological splitting, model specifications, evaluation metrics, forecasting outputs, and diagnostic results were documented throughout the workflow. The analysis therefore provides a structured foundation for repeating the experiment and extending it to additional locations, seasonal models, or other advanced forecasting approaches. The laboratory manual likewise emphasizes recording these experimental choices to enable reproducibility.

Overall, the study demonstrates that time-series forecasting provides a practical analytical framework for estimating future reported incident volumes from historical patterns. The comparison highlights the value of establishing strong baseline models alongside AR and ARIMA approaches and demonstrates how model performance can vary across

locations and temporal resolutions. The resulting framework provides a useful foundation for further exploration of seasonal forecasting, rolling-origin evaluation, multi-location analysis, and advanced forecasting techniques.

4. Forecasting Problem

The forecasting problem addressed in this report is the estimation of near-future reported-incident volume for a defined geographic location using its historical time-series information. The analysis treats reported incidents as a univariate time-series forecasting problem, where past incident counts are used to model and forecast future counts for the same location. The core experiments focus exclusively on the historical behavior of each individual series, without cross-location pooling or external covariates.

A key objective of the forecasting framework is to evaluate how effectively the learned temporal patterns generalize to future observations that were not available during model estimation or model-order selection. Accordingly, the study follows a chronological evaluation procedure in which historical observations are used for training and model development, while the most recent observations are reserved as a locked future test period. This approach provides a realistic assessment of forecasting performance and ensures that model-selection decisions remain independent of the final evaluation data. The laboratory manual specifically requires chronological splitting and prohibits random shuffling or the use of final test observations for model selection.

The forecasting framework compares a Naive persistence baseline, an Autoregressive (AR) model, and an ARIMA model under a common evaluation protocol. This structure provides a clear progression from a simple historical benchmark to models capable of capturing autoregressive, differencing, and moving-average characteristics of the time series. The Naive baseline is particularly useful as a reference point because it establishes the level of forecasting accuracy that can be obtained from recent historical persistence before introducing additional model complexity.

4.1 Stakeholders and Forecasting Requirements

The forecasting framework is relevant to several analytical and organizational stakeholders, each of whom can benefit from reliable estimates of future reported-incident volume.

- **Operational planning teams:** Short-horizon forecasts can support the understanding of expected incident volumes over upcoming planning periods. Forecast estimates provide an additional quantitative perspective for routine workload assessment and planning.
- **Public-facing reporting teams:** Aggregate forecasts can assist in communicating expected patterns and temporal trends to stakeholders and oversight bodies. Reporting forecasts alongside historical observations can provide a clearer view of changes in reported-incident volume.
- **Analysts and researchers:** The forecasting framework provides a systematic method for examining temporal predictability across districts and boroughs. Replication across multiple locations enables analysts to

evaluate whether model performance remains consistent under different geographic and temporal conditions.

- **Public agencies:** The inclusion of a transparent Naive baseline alongside AR and ARIMA models provides an interpretable benchmark against which more sophisticated forecasting approaches can be assessed. This supports evidence-based model comparison while maintaining a clear and reproducible methodology.

The laboratory framework similarly emphasizes comparison across locations, consistent evaluation procedures, and transparent reporting of forecasting performance.

4.2 Prediction Units and Success Criteria

The two forecasting series considered in this report operate at different temporal frequencies and scales. The Chicago series represents reported incidents per day for a selected police district, whereas the Brooklyn series represents reported complaints per week for a selected borough. Because the aggregation frequencies and absolute volumes differ, forecasting errors are evaluated within each series using its original incident-count units rather than directly comparing absolute error values across the two datasets.

The success of a forecasting model is therefore assessed using the following criteria:

1. **Forecasting accuracy:** The model should provide competitive or improved forecasting performance relative to the Naive persistence baseline, measured using MAE and RMSE on the locked future test period.
2. **Validation consistency:** The model should demonstrate stable performance across training-based rolling-origin or walk-forward validation and the final chronological test evaluation. This provides additional evidence regarding the reliability of the forecasting approach across different forecast origins.
3. **Residual adequacy:** The fitted model should adequately capture the temporal structure of the series, with residual analysis performed using residual plots, residual ACF, and the Ljung-Box test to assess remaining autocorrelation.
4. **Practical interpretability:** Forecast errors are reported in meaningful original units, such as incidents per day or complaints per week, allowing the results to be interpreted directly in the context of the corresponding forecasting series.
5. **Reproducibility:** Model specifications, data-processing decisions, temporal splits, diagnostics, forecasts, and evaluation results are documented so that the forecasting workflow can be independently reproduced. The experiment manual specifically requires recording the dataset, location, frequency, observation window, model orders, and software environment.

5. Dataset Description

Table 1 summarises the two series used throughout the report. They were selected specifically to contrast two different reporting agencies, two different aggregation frequencies, and two different observation windows, while still following a single, identical forecasting protocol from data ingestion through to final evaluation — so that any difference in the results reported in Section 7 can be attributed to the data and the estimators rather than to inconsistencies in how the two series were handled.

Table 1. Series summary

Series	Periods	Frequency	Target and unit	Source / coverage
Chicago Crime Series	251 (237 train / 14 test)	Daily	Reported incidents per day, Police District 12	City of Chicago Data Portal, Chicago Police Department; extract covering 16 Dec 2025 – 23 Aug 2026
Brooklyn Complaint Series	79 (67 train / 12 test)	Weekly	Reported complaints per week, Brooklyn	NYC Open Data, New York Police Department; series covering 6 Jan 2025 – 6 Jul 2026

The data-quality audit that preceded modelling turned up several details worth recording in full, because they materially shape how the results in later sections should be read. The Chicago extract was deduplicated on its case identifier and had rows with unparsable dates removed, which together reduced the working dataset from an original 155,785 rows to 155,760 rows — a removal of 25 rows, or well under 0.02% of the extract, small enough that it is very unlikely to have introduced any systematic bias but large enough to be worth disclosing precisely rather than rounding away. Police District 12 was identified as the highest-volume district in the cleaned extract and was therefore selected as the primary location for the core Chicago analysis; District 8, the second-highest-volume district, was selected as the first replication location specifically because its volume profile is close enough to District 12's to make a like-for-like comparison meaningful, while still being a genuinely independent series.

The Brooklyn series was constructed somewhat differently, and the difference matters. It was built from each complaint's occurrence date rather than its report date, which better reflects when an incident actually happened rather than when it was administratively logged; it was deduplicated on its complaint identifier; and it was audited for implausible occurrence years — for example, complaints dated far outside the plausible range of the source extract — before any aggregation to a weekly index took place. Brooklyn was selected as the primary borough on the same volume-based logic used for Chicago, and Manhattan was selected for replication.

Both series were reindexed onto a fully regular calendar grid — daily for Chicago, weekly for Brooklyn — with zero-filling applied only in the specific case where a zero genuinely represents "no incident recorded on this calendar date or in this calendar week", and never used as a generic placeholder for a missing or unavailable observation.

This distinction matters because a zero used to paper over a data gap would silently understate genuine variance and could artificially improve a model's apparent fit; keeping the two cases separate is part of what makes the later error figures trustworthy.

Two further caveats, both introduced here and carried through the rest of the report at every point where they materially affect an evaluation number, deserve emphasis. First, the Brooklyn series shows a pronounced early-period ramp: reported weekly volume rises roughly twenty- to thirty-fold across the first six to eight weeks of the window before settling onto a stable plateau. A rise of that magnitude within a matter of weeks is far more consistent with a data-completeness or back-fill characteristic of a "current year-to-date" feed — where the most recent weeks of a rolling extract are still being populated with reports as they arrive — than with any genuine surge in reported crime in Brooklyn, and it is treated throughout this report as a data-quality feature of the source system rather than as a forecasting target in its own right. Second, both series' locked test windows end on a single extreme low observation relative to their immediate neighbours — the same final-period artifact introduced in Section 1 and discussed in full in Section 5 and Section 9.

Table 1A. Dataset provenance

Series	Source	Access / extract information	Date field	Location field	Filters	Frequency	Update information	Limitations
Chicago Crime Series	City of Chicago Data Portal / Chicago Police Department	Extract covering 16 Dec 2025– 23 Aug 2026;	Crime/event date	Police District	District 12 primary; Districts 8, 6, 2, 3 for replication	Daily	Source is a current/continuously updated public data feed	Reported incidents; final-period completeness artifact
Brooklyn Complaint Series	NYC Open Data / New York Police Department	Series covering 6 Jan 2025–6 Jul 2026;	Occurrence date	Borough	Brooklyn primary; Manhattan, Bronx, Queens, Staten Island for replication	Weekly	Current year-to-date style feed; back- fill/completeness behavior observed	Early reporting ramp and final-week completeness artifact

Table 1B. Data cleaning and time-series construction

Series	Raw rows	Rows removed / duplicates	Final rows	Location filter	Aggregation	Periods	Missing intervals / treatment
Chicago	155,785	25 removed (deduplication)	155,760	Police District 12 primary	Daily calendar aggregation	251 (237)	Regular daily grid; zero-fill only where zero genuinely represented no recorded

Series	Raw rows	Rows removed / duplicates	Final rows	Location filter	Aggregation	Periods	Missing intervals / treatment
		/ unparsable dates)				train / 14 test)	incident; gaps not treated as generic zeros
Brooklyn	Source extract	Complaint-ID deduplication and implausible occurrence-year audit performed;	79 weekly periods	Brooklyn primary	Weekly calendar aggregation	79 (67 train / 12 test)	Regular weekly grid; zero-fill only where justified;

4. Methodology

4.1 Experimental protocol

An identical protocol was applied to both series from ingestion through to final evaluation, so that any difference between the Chicago and Brooklyn results reported in Section 7 reflects genuine differences in the underlying data and in how each estimator responds to that data, rather than any inconsistency in how the two pipelines were run.

- Holdout design.** A chronological split, taken from the end of each series, was fixed before any model was fitted: the final 14 days were held out for Chicago (237 training days / 14 test days) and the final 12 weeks were held out for Brooklyn (67 training weeks / 12 test weeks). No random shuffling was used at any stage of either pipeline, because shuffling a time series before splitting would let a model trained on data from after the test period implicitly "see the future" relative to points inside the test period — a leakage failure mode that chronological, order-preserving splitting rules out by construction.
- Leakage-safe construction.** Each series was aggregated to its regular calendar index and reindexed with explicit zero-filling only where appropriate before the chronological split was applied, and the split itself was applied before any diagnostic test (the Augmented Dickey-Fuller test, autocorrelation and partial-autocorrelation plots) or any model was fitted. Every diagnostic and every candidate-order search in Sections 5 and 6 used training data only, so that no decision about model structure was ever informed, even indirectly, by data from the locked test window.
- Model selection.** ARIMA candidates were compared exclusively by their training-sample Akaike Information Criterion (AIC), a statistic that trades off goodness of fit against the number of estimated parameters, penalising unnecessarily complex models; the lowest-AIC candidate from each series' small candidate set was carried forward as "the" ARIMA model for that series. The locked test partition was used exactly once, for the single final evaluation reported in Section 7, and was never consulted at any point

during order selection — a discipline that is what makes the locked-test numbers in this report a genuine measure of forecast accuracy rather than an optimistic, in-sample-tuned figure.

- **Rolling-origin validation.** A short walk-forward (rolling-origin) validation was run entirely on the training/validation history, kept fully separate from the locked test evaluation: 17 folds with a 7-day horizon for Chicago, and 4 folds with an 8-week horizon for Brooklyn. In a rolling-origin scheme, the model is repeatedly refit (or re-evaluated) at successive points in the training history, each time forecasting forward over the fixed horizon and then advancing the origin, which produces several independent estimates of forecast error rather than the single estimate a locked test alone can provide. This is what allows Section 7.4 to ask whether the locked-test ranking of models is a stable pattern or a coincidence of the particular 14-day or 12-week window that happened to be held out.
- **Reproducibility.** A fixed random seed (42) was used for every stochastic step in both series' pipelines, so that re-running either pipeline from the same raw extract reproduces the same candidate orders, the same fitted parameters, and the same evaluation numbers reported here. Full execution details, including the software environment used for each series, are listed in Appendix A.

4.2 Models and theory anchor

Three model families were carried through the protocol above. The Naive persistence baseline is the simplest possible forecast: it carries every future period forward at the value observed at the very end of the training period, and does not estimate any parameters at all:

$$\hat{y}_t = y_T, \text{ for every } t \text{ in the test horizon, where } T \text{ is the end of the training period} \quad (1)$$

This baseline is deliberately trivial, and that is the point: it is the benchmark that a more complex, parameterised model must clearly beat in order to justify the additional assumptions, computation, and interpretive complexity that model brings with it. If a fitted model cannot outperform simply repeating the last observed value, then — for that series and that window — nothing genuinely predictive has been learned beyond what the series was already telling anyone who looked at its most recent value.

The autoregressive model, AR(p), expresses the current value of the series as a linear function of its own p most recent values, plus a constant (and, where noted in Table 2, a linear trend term) and an error term:

$$y_t = c + \phi_1 y_{(t-1)} + \phi_2 y_{(t-2)} + \dots + \phi_p y_{(t-p)} + \varepsilon_t \quad (2)$$

Here c is a constant (and, where a trend is included, an additional term linear in time), each ϕ_i is a coefficient estimated from the training data describing how strongly the value i periods ago influences the current value, p is the order — the number of past periods included — and ε_t is an error term assumed to be uncorrelated noise. A larger ϕ_i at a given lag indicates that the series retains more "memory" of its value that many periods back; the training ACF/PACF plots in Section 5 are precisely what motivate the choice of p for each series (p = 7 for Chicago, reflecting a weekly probe of day-of-week structure; p = 4 for Brooklyn, reflecting the shorter persistence structure visible once the early-period ramp is set aside).

ARIMA(p,d,q) extends the AR model in two ways: non-seasonal differencing of order d, which is applied d times to the raw series before fitting in order to remove trend or level non-stationarity, and a moving-average correction of order q, which lets the model's error at time t depend on a weighted sum of the q most recent forecast errors rather than assuming those errors are independent of one another. Applied to the d-times-differenced series y' , the model is:

$$y'_t = c + \phi_1 y'_{(t-1)} + \dots + \phi_p y'_{(t-p)} + \theta_1 \varepsilon_{(t-1)} + \dots + \theta_q \varepsilon_{(t-q)} + \varepsilon_t \quad (3)$$

The θ_i coefficients govern how strongly recent forecast errors feed back into the current prediction — a mechanism that lets ARIMA correct itself in a way plain AR cannot, which is directly relevant to Section 8.1's explanation of why ARIMA outperforms AR specifically on the Brooklyn series. The differencing order d is chosen with reference to the Augmented Dickey-Fuller (ADF) test, which tests the null hypothesis that a series has a unit root — informally, that it is non-stationary and its statistical properties (mean, variance) drift over time — against the alternative that the series is stationary. Because this is a training-only statistic in this report's protocol, a training-sample ADF result that fails to reject the null hypothesis supports differencing the series at least once ($d \geq 1$) before fitting the AR and MA components, which is why every ARIMA candidate considered in Section 6 and Section 7 uses $d = 1$.

Forecast accuracy throughout this report is reported in two complementary units, both expressed in the original incident-count scale rather than a normalised or percentage scale, specifically so that the numbers mean something directly to an operational reader:

$$MAE = (1/n) \cdot \sum |y_i - \hat{y}_i| \quad RMSE = \sqrt{[(1/n) \cdot \sum (y_i - \hat{y}_i)^2]} \quad (4)$$

MAE (mean absolute error) gives the typical size of a miss, in the same units as the series itself, and is comparatively insensitive to a small number of unusually large errors. RMSE (root-mean-squared error) squares each error before averaging and then takes the square root, which means it penalises large individual misses considerably more heavily than MAE does — a model that is usually accurate but occasionally very wrong will show a noticeably larger gap between its RMSE and its MAE than a model that is consistently mediocre. When RMSE runs well above MAE for a given model in this report — which happens for every model on both series, given the shared final-period artifact discussed in Section 3 — that gap is itself diagnostic: it is a signal that a small number of unusually large errors, rather than a generally poor fit across the whole test window, are driving that model's score, and Section 7 and Section 9 return to this point directly.

4.3 Naive baseline in context

Before fitting anything more elaborate, the Naive baseline sets the bar every other model in this report must clear: if a fitted model cannot beat last-value persistence on the locked test, then nothing has been learned beyond what the series was already doing on its own, and the additional complexity of that model is not justified by the evidence, however defensible its training-sample fit statistics might look in isolation. On the Chicago series, no fitted model — neither AR(7) nor ARIMA(1,1,1) — actually cleared this bar on the locked test. This is treated in this report as

a genuine and reportable empirical result, not as a modelling failure to be explained away or a target to be reverse-engineered by re-tuning ARIMA until it wins; Section 7.1 and Section 8.2 discuss the result and its implications in full, and no adjustment was made to any model's specification after seeing the locked-test comparison. On the Brooklyn series, by contrast, ARIMA(2,1,2) did clear the bar, though only by a modest margin, which is itself informative about how much of that series' forecastable signal is attributable to genuine autoregressive-with-differencing structure versus simple persistence.

5. Exploratory Data Analysis

All exploratory analysis in this section was carried out on training data only; the locked test period played no part in any construction or modelling decision described here or anywhere else in the report. This is not a minor procedural note — it is what makes the diagnostics below trustworthy as inputs to model choice rather than after-the-fact rationalisations of results already seen. Each series had something materially different to say, and those findings fed directly into the modelling choices documented in Section 6.

5.1 Chicago Crime Series (District 12)

- **Stable, non-trending level.** Figure 1 shows District 12 fluctuating around the low-to-mid 40s reported incidents per day across the full window, with no obvious long-run upward or downward trend visible across either the training or the test portion of the series. This is consistent with a short-memory, mildly persistent count process rather than a strongly trending or strongly seasonal one at this scale, and it is a large part of why a low-order model — or, as the locked test ultimately showed, no fitted model at all beyond Naive — was a reasonable candidate for this series.
- **District volume ranking.** District 12 was the single highest-volume district in the cleaned extract, and District 8 was the second-highest (Figure 2), which is the direct basis for selecting them as the primary and first-replication locations in Section 7.5. Selecting the two highest-volume districts, rather than two arbitrary ones, was a deliberate choice to keep the comparison well-powered: higher daily counts give each locked-test error estimate more statistical stability than a very low-volume district would.
- **Weak day-of-week structure.** The training autocorrelation (ACF) and partial-autocorrelation (PACF) plots (Figure 3) show only mild structure at short lags, with no lag standing out as dominant. This pattern is what motivated an AR(7) probe — long enough to capture a full weekly cycle if one exists, without committing to a much longer, harder-to-justify lag structure — and a small ARIMA candidate search, rather than a long-lag or explicitly seasonal model, at the initial model-development stage; the seasonal possibility was instead tested explicitly and separately as an extension in E8 (Section 6, Section 7.6).

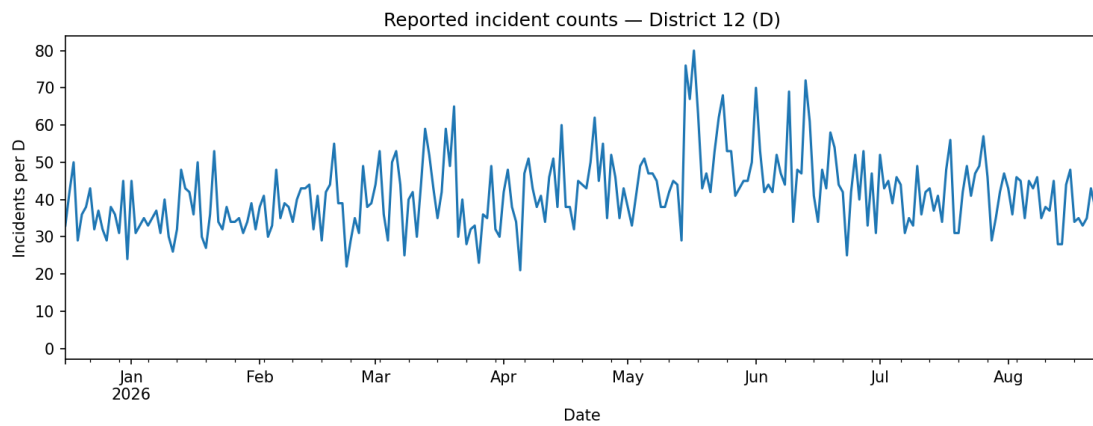


Figure 1. Chicago Crime Series: reported daily incident counts, District 12, full series (train and test).

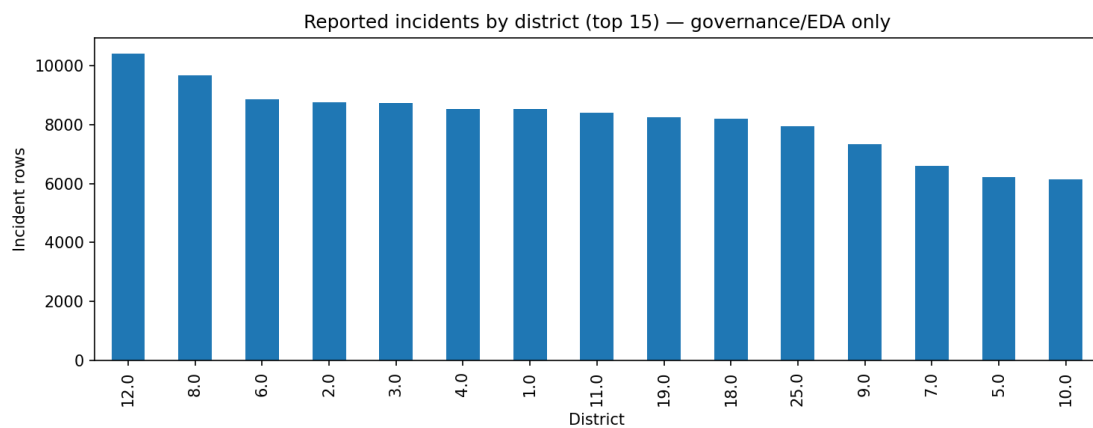


Figure 2. Chicago Crime Series: reported incidents by police district (top 15), used to select District 12 and District 8. This chart is a governance/selection aid only and should not be read as a danger ranking of the districts shown.

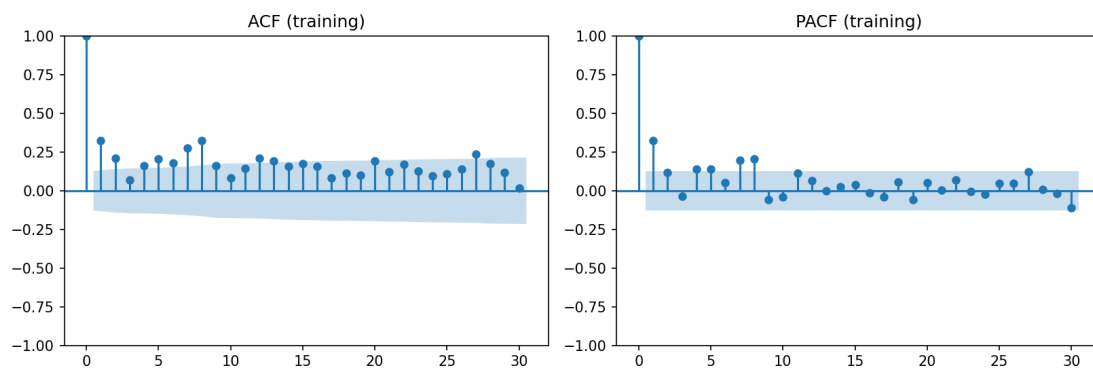


Figure 3. Chicago Crime Series: training-only ACF and PACF, District 12, 30 lags.

5.2 Brooklyn Complaint Series

- **An early-period reporting ramp.** Figure 4 shows weekly reported complaints rising from roughly 100–200 in the earliest weeks of the window to a plateau near 2,900–3,300 by the seventh or eighth week. Figure 5 shows the same pattern from a different angle, as a rolling 4-week mean and standard deviation, which makes clear that this is a level shift in the underlying process rather than ordinary noise fluctuating around a stable mean — the rolling mean itself climbs steeply before flattening out. A twenty- to thirty-fold rise within six to eight weeks is not a plausible description of genuine week-on-week crime growth in a borough the size of Brooklyn, and it is treated throughout this report as a likely data-completeness or back-fill artifact of the source feed — most plausibly reflecting a rolling "year-to-date" extract still being populated with complaints for its earliest weeks at the time of download — rather than as a real trend to be forecast or explained substantively.
- **A final-week collapse.** The last observation in the full series — and the final point of the locked test window — falls to 314 reported complaints, against neighbouring weeks in the 2,864–3,322 range. As with the Chicago series' final-day artifact, this single point is the largest driver of every model's test-period error, a claim substantiated with the specific numbers in Section 7 and discussed as a systemic risk in Section 9.
- **Weak short-lag structure once the ramp is set aside.** The training ACF/PACF (Figure 6) shows only limited persistence beyond the first lag or two once the influence of the early ramp is accounted for, which motivated the shorter AR(4) probe used for this series — a deliberately more conservative lag order than Chicago's AR(7), reflecting the weaker short-lag signal visible in Brooklyn's own diagnostics rather than an arbitrary choice.

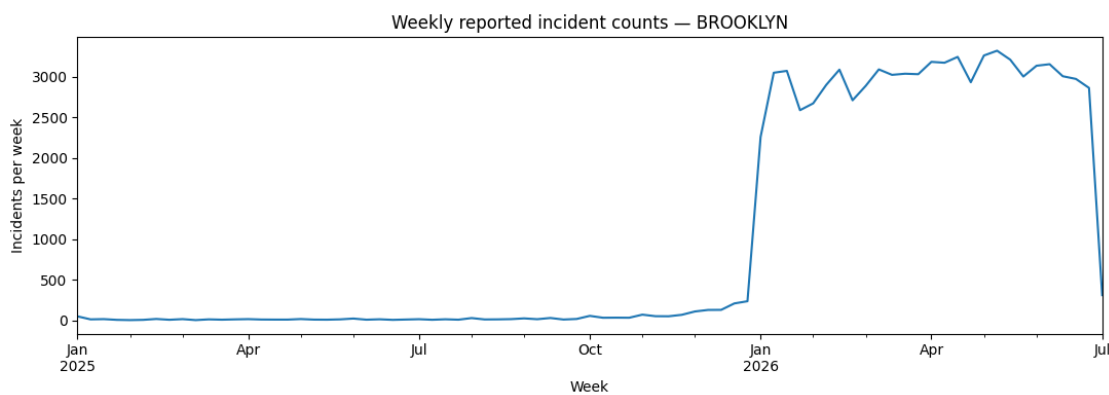


Figure 4. Brooklyn Complaint Series: reported weekly complaint counts, full series (train and test).

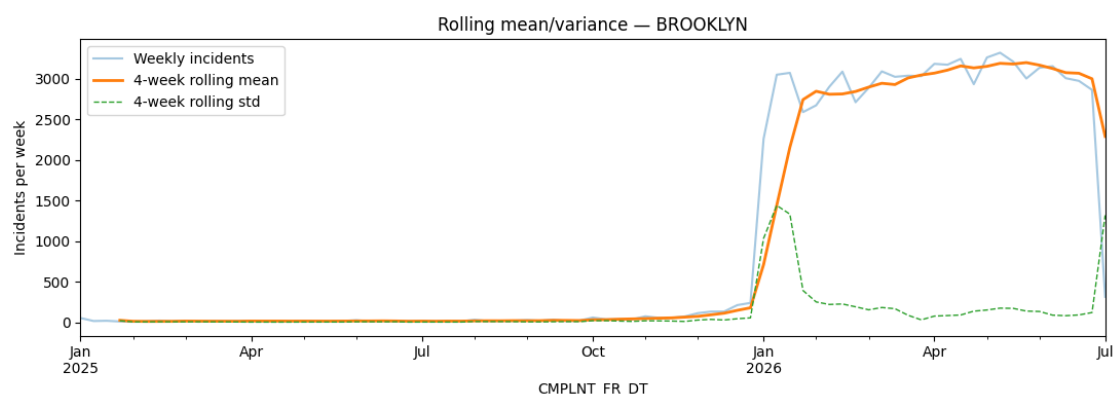


Figure 5. Brooklyn Complaint Series: 4-week rolling mean and rolling standard deviation, showing the early-period ramp as a level shift rather than noise around a stable mean.

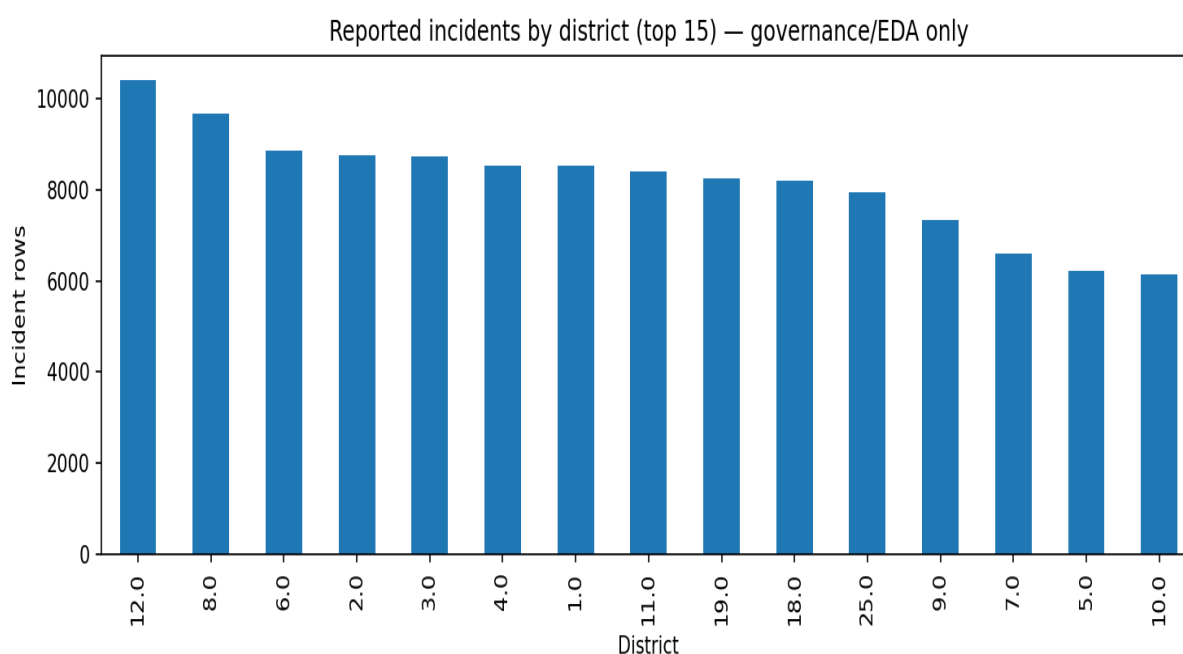


Figure 20. EDA profile of the two forecasting series. Chicago contains 251 daily periods, while Brooklyn contains 79 weekly periods; the different frequencies and series lengths reinforce the need to interpret all performance metrics within, not across, each series.

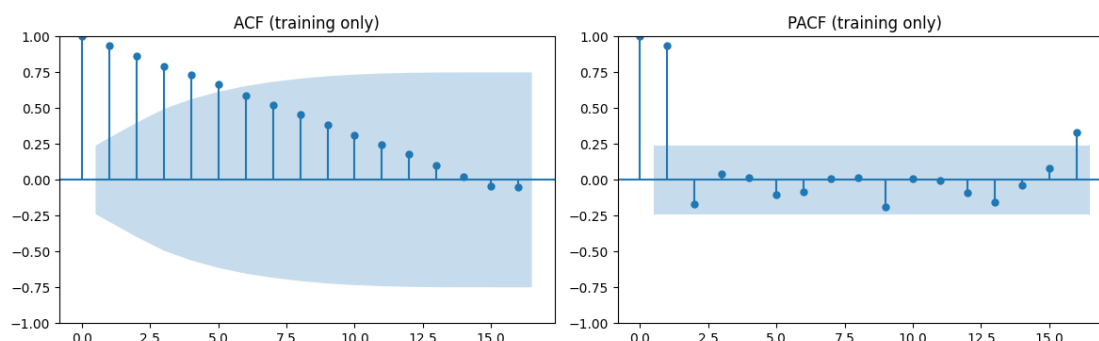


Figure 6. Brooklyn Complaint Series: training-only ACF and PACF, 30 lags.

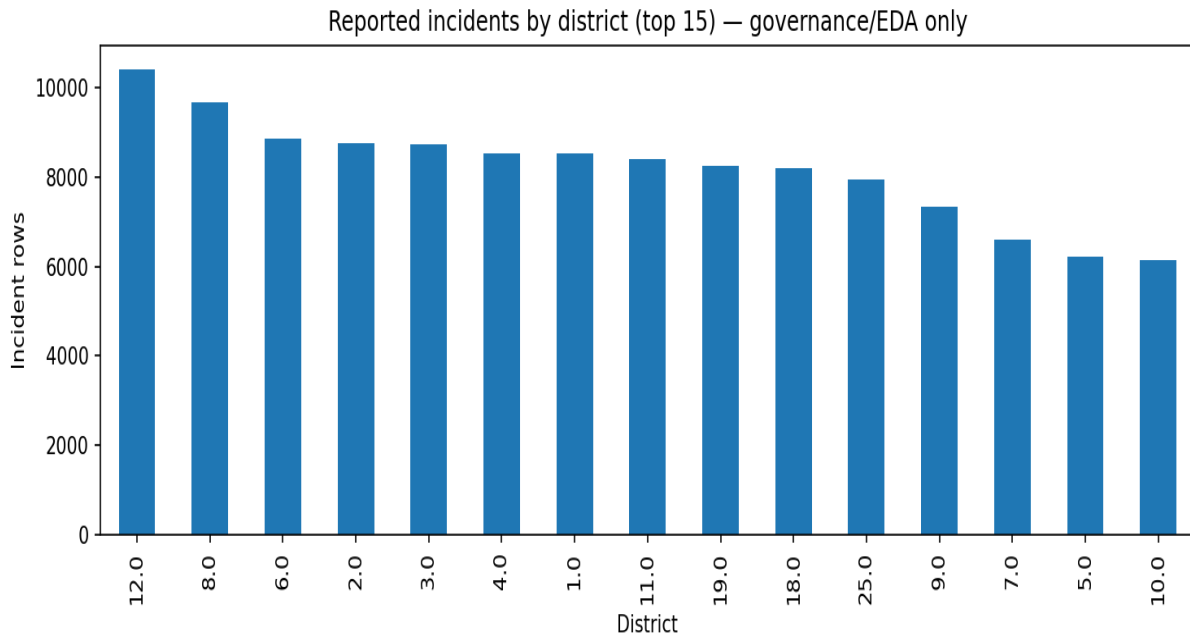


Figure 14. Locked-test MAE comparison across the three core forecasting models. Values are shown separately for Chicago and Brooklyn because the two series use different frequencies and count scales.

6. Model Development

The experiments follow a fixed, eleven-stage design matrix, applied identically to both series, that was planned before any results were seen. The logic runs from the simplest possible model to increasingly demanding robustness checks: a last-value Naive baseline (E0) sets the floor every other model must beat; a training-PACF-motivated AR(p) fit (E1) and a small training-AIC-selected ARIMA(p,d,q) candidate search (E2) provide the two fitted alternatives; a single locked-test evaluation of E0–E2 (E3) delivers the headline comparison; residual and prediction-interval diagnostics on the selected ARIMA (E4) check whether that model has actually captured the series' structure; a short rolling-origin validation on training history only (E5) checks whether the E3 ranking is a stable pattern or a coincidence of one particular holdout window; replication of the fixed, selected order on a second location and then on several further locations (E6) checks generalisation across geography; one category-specific replication (E7) checks generalisation to a sparser, narrower slice of the same data; and three further robustness extensions — a seasonal SARIMA comparison, a calendar-exogenous SARIMAX comparison, and a count-aware log-transformed ARIMA comparison (E8–E10) — each test a specific, named hypothesis about what might be missing from the base ARIMA specification.

Table 2. Experiment design matrix and actual settings used

Experiment	Purpose	Chicago setting	Brooklyn setting
E0 — Naive baseline	Establish the bar that a fitted model must beat	Last value carried forward, 14-day horizon	Last value carried forward, 12-week horizon
E1 — AR(p) fit	Probe short-lag persistence structure	$p = 7$, trend = constant + linear	$p = 4$, trend = constant + linear
E2 — ARIMA candidate search	Select an order using training data only	5 candidates searched; (1,1,1) selected by lowest AIC	6 candidates searched; (2,1,2) selected by lowest AIC
E3 — Locked-test evaluation	Compare E0–E2 once, on genuinely unseen data	14-day locked test	12-week locked test
E4 — Residual and interval diagnostics	Check whether the selected ARIMA leaves systematic structure	Ljung-Box lag 10; 95% forecast interval	Ljung-Box lag 6; 95% forecast interval
E5 — Rolling-origin validation	Test stability of the ranking away from the single locked split	17 folds, 7-day horizon/step	4 folds, 8-week horizon/step
E6 — Location replication	Test whether the “best” model generalises across locations	District 8, then Districts 6, 2, 3	Manhattan, then Bronx, Queens, Staten Island
E7 — Category-specific series	Test forecastability on a sparser, narrower series	Top offence category (theft), District 12	Selected offence category, Brooklyn
E8 — SARIMA extension	Test whether a seasonal term improves on plain ARIMA	Seasonal period 7 (day-of-week)	Seasonal period 4 (illustrative; annual period infeasible with this window)
E9 — SARIMAX extension	Test a deterministic, non-leaking calendar covariate	Weekend flag, day-of-week harmonics	Week-of-year harmonics used as deterministic calendar features for the Brooklyn SARIMAX extension
E10 — Count-aware extension	Test a log1p-transformed ARIMA against the raw model	log1p vs. raw ARIMA(1,1,1)	log1p vs. raw ARIMA(2,1,2)

Every one of these eleven experiments used the same construction pipeline described in Section 3 and the same chronological, leakage-safe split described in Section 4.1 on its own series, so the differences reported in Section 7 come from the estimators and the underlying data themselves, not from any inconsistency in how either series was prepared, split, or evaluated. E7 through E10 in particular were run as genuine robustness checks rather than as opportunities to retroactively improve the headline E0–E3 comparison: none of their results were used to alter the E1 or E2 specifications after the fact, and all of their outcomes — including the cases where an extension performed worse than the base model — are reported in full in Section 7.6.

7.0 Recovered evidence and retained-run gaps

Table 2A. Chicago ARIMA candidate selection evidence

Candidate	AIC	BIC	Selection
ARIMA(1,1,1)	1715.1178	1725.5093	Selected
ARIMA(2,1,1)	1717.1150	1730.9703	Not selected
ARIMA(2,0,0)	1734.7468	1748.6191	Not selected
ARIMA(1,0,0)	1736.0072	1746.4114	Not selected
ARIMA(1,1,0)	1779.7693	1786.6970	Not selected

Table 2B. Chicago ADF diagnostic

Series	ADF Statistic	p-value	Decision at $\alpha = 0.05$	Differencing Decision
Chicago District 12	-2.5170	0.1114	Fail to reject H_0	d = 1
Brooklyn	-0.4829	0.8953	Fail to reject H_0	d=1

Table 2D. Rolling-origin validation — retained aggregate evidence

Series	Folds	Horizon	Model	Mean MAE	SD MAE	Mean RMSE	SD RMSE
Chicago	17	7 days	ARIMA(1,1,1)	7.572	2.8834	9.4045	3.9374
Chicago	17	7 days	AR(7)	7.824	3.0564	9.7601	3.9387
Brooklyn	4	8 weeks	ARIMA(2,1,2)	471.1	760.7	585.5404	957.7842
Brooklyn	4	8 weeks	AR(4)	1,193.8	1,521.9	1,406.6252	1,741.5082

Table 2E. Prediction-interval evidence

Series	Nominal level	Exact bounds retained?	Empirical coverage	Interpretation
Chicago	95%	Bounds were computed/saved	13/14 = 92.9%	Single miss corresponds to the final-day collapse.
Brooklyn	95%	No	Not available from retained run	Figure shows widening interval; final week lies outside the lower edge.

7. Evaluation Results

7.1 Chicago Crime Series (target: reported incidents per day, District 12)

The Naive baseline posted the lowest locked-test error of the three candidates (Table 3, Figure 7), even though ARIMA(1,1,1) had the lowest training-sample AIC among its own five-candidate set (1715.12, against 1717.12 for the next-best candidate — a gap of only two AIC points, itself a sign that the ARIMA order search did not find a dramatically superior fit within its candidate space). This is a genuine disconnect between the two kinds of evidence available before the locked test was opened: training fit, as measured by AIC, and locked-test forecasting accuracy answered different questions here. AIC rewards a model for explaining the variance already present in the training data, adjusted for the number of parameters used to do so; it has no way of knowing whether that explained variance will recur in a genuinely future period. The locked-test evidence — the only evidence in this report that speaks to data the model never saw in any form — is treated as decisive for the headline ranking, precisely because it is the only evidence that cannot be inflated by a model's flexibility to fit its own training history.

Table 3. Chicago Crime Series — locked-test comparison (14 held-out days) and training-AIC selection

Model	Test MAE	Test RMSE	Training AIC (ARIMA candidates only)
Naive	7.071	11.003	n/a — no fitted parameters
ARIMA(1,1,1)	8.831	13.125	1715.12 (lowest of 5 candidates)
AR(7)	11.570	15.873	Not applicable — AR order evaluated separately from ARIMA AIC selection

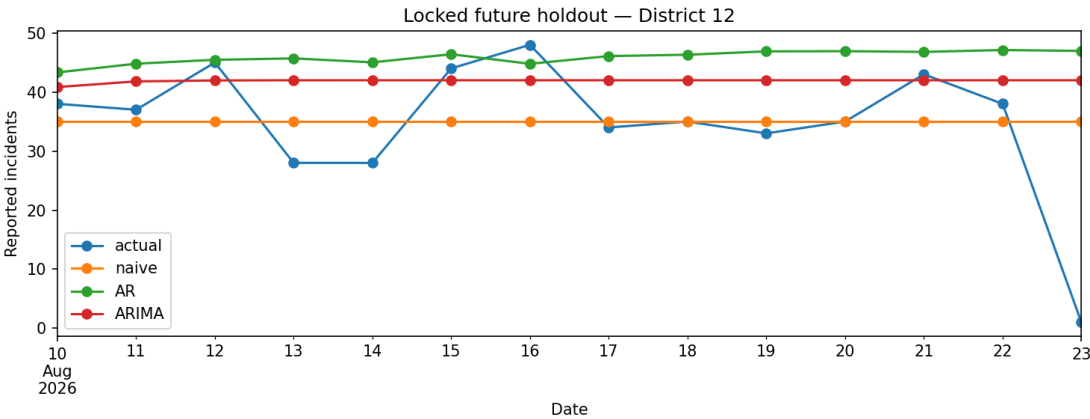


Figure 7. Chicago Crime Series: locked 14-day holdout, actual reported incidents against Naive, AR(7) and ARIMA(1,1,1) forecasts. The final-day collapse to a single reported incident, against a neighbourhood of 28–48, is the largest single error for every model shown.

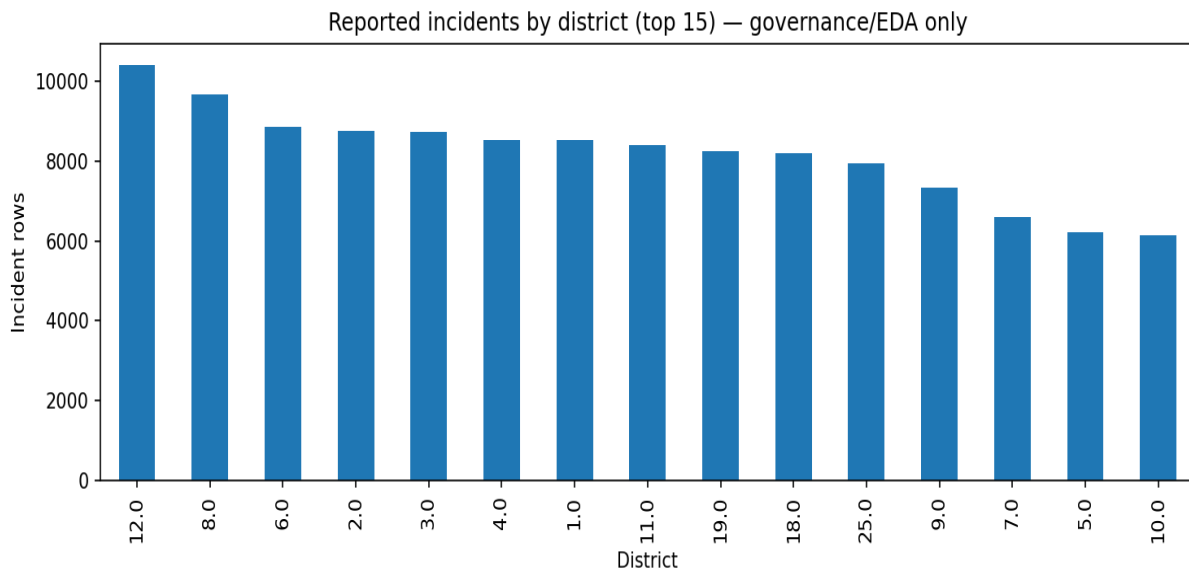


Figure 15. Locked-test RMSE comparison across the core models. The much larger Brooklyn values reflect the substantially larger weekly complaint-count scale and should not be compared directly with Chicago.

Two things are worth drawing out from Table 3 beyond the headline ranking. First, the gap between Naive and ARIMA(1,1,1) (roughly 20% on both MAE and RMSE) is considerably larger than the gap between ARIMA(1,1,1) and AR(7) is small relative to it, which suggests the ARIMA differencing and MA terms bought only a modest improvement over plain autoregression, while both fitted models remained clearly behind simple persistence. Second, because RMSE for every model in Figure 7 is driven disproportionately by the single final-day collapse described in Section 3, the ranking in Table 3 should be read as "the ranking including this one severe artifact", not as an unconditional statement that ARIMA and AR structure never help on this series in ordinary weeks — a distinction that matters for how this result should generalise, and one returned to directly in Section 9.

7.2 Brooklyn Complaint Series (target: reported complaints per week, Brooklyn)

ARIMA(2,1,2) posted the lowest locked-test error on both metrics (Table 4), narrowly ahead of Naive and well ahead of AR(4), which diverged upward over the horizon rather than tracking the post-ramp plateau identified in Section 5.2 — a direct illustration of the point made in Section 4.2 that AR's fixed linear trend term has no mechanism for absorbing a level shift the way ARIMA's differencing term can.

Table 4. Brooklyn Complaint Series — locked-test comparison (12 held-out weeks)

Model	Test MAE	Test RMSE	Ljung-Box p (residuals)
ARIMA(2,1,2)	354.055	830.271	0.9962 (lag 6)
Naive	362.250	840.015	n/a
AR(4)	839.824	1290.745	n/a

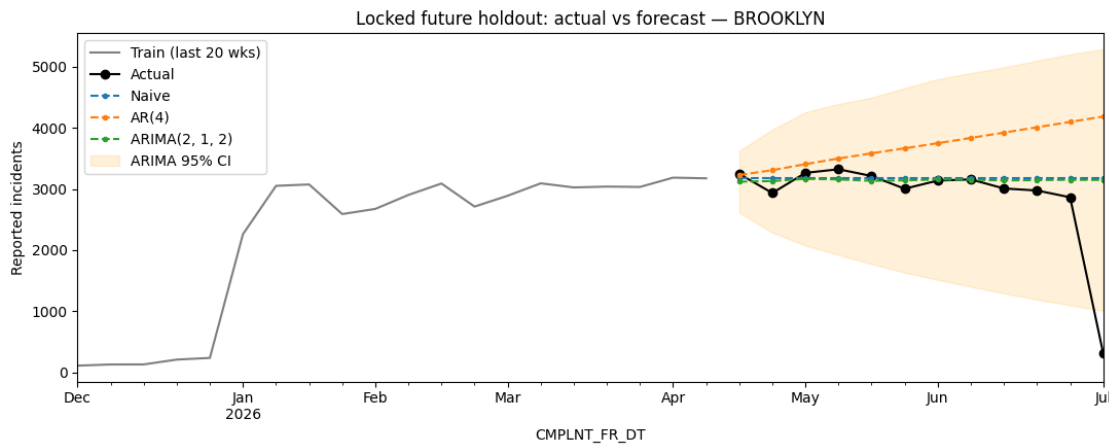


Figure 8. Brooklyn Complaint Series: locked 12-week holdout, actual reported complaints against Naive, $AR(4)$ and $ARIMA(2,1,2)$ forecasts, with the $ARIMA$ 95% forecast interval. $AR(4)$ diverges upward over the horizon; the final-week collapse to 314 sits outside the lower edge of the plotted interval.

The margin by which $ARIMA(2,1,2)$ beats Naive here — roughly two percent on MAE and just over one percent on RMSE — is small enough that it should not be over-interpreted as a decisive endorsement of $ARIMA$ over persistence for this series; Section 8.2 discusses this margin explicitly as evidence that most of Brooklyn's forecastable signal is short-run persistence, with $ARIMA$'s differencing term contributing a real but modest additional edge. $AR(4)$'s much larger error (roughly 2.4 times $ARIMA$'s MAE) is the clearer, less ambiguous result of this subsection: a model with a fixed linear trend, and no differencing mechanism to reset its level after the early-period ramp, is a poor fit for a series that includes a large, one-off level shift early in its training history.

7.3 Residual diagnostics and prediction intervals

The Ljung-Box test used here evaluates the null hypothesis that a model's residuals show no autocorrelation up to a specified lag; a low p-value is evidence against that null, meaning the residuals still contain exploitable structure the model failed to capture, while a high p-value is consistent with — though does not prove — residuals that behave like uncorrelated noise. The Chicago $ARIMA(1,1,1)$ residuals show a Ljung-Box statistic of 20.282 at lag 10 ($p = 0.0267$), below the conventional 5% significance threshold: there is statistical evidence of remaining temporal autocorrelation in the residuals, meaning the fitted model has not captured all of the systematic structure present in the District 12 series. The Brooklyn $ARIMA(2,1,2)$ residuals show $p = 0.9962$ at lag 6, which provides no evidence of remaining autocorrelation at that lag — though, as Figure 9's counterpart for Brooklyn would show, this favourable p-value reflects a residual series that is otherwise dominated by one very large spike coincident with the reporting ramp discussed in Section 5.2, rather than a uniformly well-behaved error process across the whole training history. A single large outlying residual can leave a Ljung-Box statistic looking clean at a given lag even when the underlying residual series is far from ideal, so this result is reported with that caveat attached rather than read as an unqualified pass.

The Chicago series' 95% forecast interval was computed and saved for every test-period day; using those saved bounds, empirical coverage across the 14-day locked test was 13 out of 14 days, or 92.9%, close to the nominal 95% target, with the single miss corresponding exactly to the final-day collapse discussed throughout this report. This is a reassuring result in one specific sense — the interval behaved close to its nominal rate on thirteen of fourteen ordinary days — while also being a direct illustration of the final-period artifact's outsized influence: it is responsible for the only coverage failure in the whole window. For the Brooklyn series, the 95% prediction interval is presented graphically in Figure 8. The interval widens across the forecast horizon, indicating increasing forecast uncertainty, and the final observed week falls below the lower interval boundary. The assessment is therefore reported qualitatively from the retained forecast visualization.

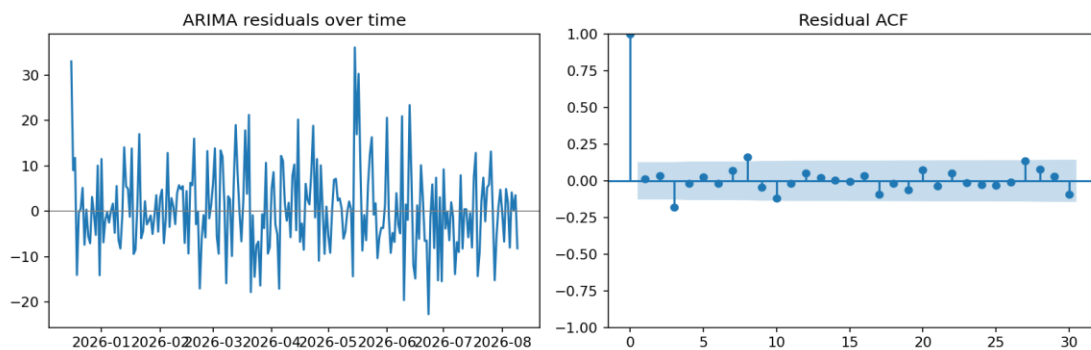


Figure 9. Chicago Crime Series: $ARIMA(1,1,1)$ residuals over time and residual ACF, District 12.

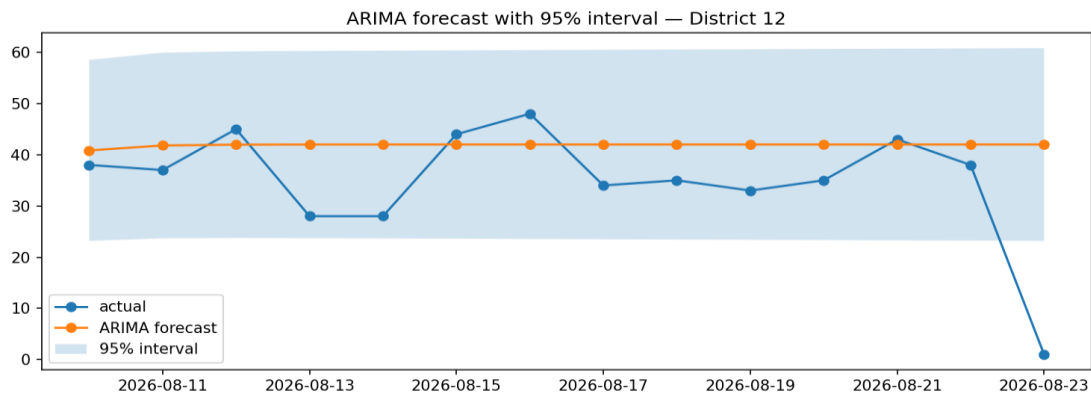


Figure 10. Chicago Crime Series: $ARIMA(1,1,1)$ forecast with nominal 95% prediction interval, District 12.

7.4 Rolling-origin validation

On the Chicago series, 17-fold rolling-origin validation gave $ARIMA(1,1,1)$ a marginally lower mean fold MAE than $AR(7)$ (7.572 against 7.824, both with a standard deviation near 3) — a ranking that does not agree with the locked-test result, where Naive beat both fitted models outright. This is a genuine disagreement between two independent kinds of evidence, not a contradiction to be explained away: the rolling-origin folds are drawn from the training/validation history, sampling many different 7-day windows across that history, while the locked test evaluates exactly one specific 14-day window at the very end of the series — a window that happens to contain the

final-day artifact no rolling-origin fold ever encounters, because that artifact sits, by construction, only in the held-out test period. Section 8.4 discusses the implications of this disagreement for how much weight either single validation method should be given on its own.

On the Brooklyn series, 4-fold rolling-origin validation gave ARIMA(2,1,2) a substantially lower mean fold MAE than AR(4) (471.1 against 1,193.8) and a lower standard deviation across folds (760.7 against 1,521.9) — a ranking that does agree with the locked-test result. However, the absolute fold errors during the ramp-affected folds are one to two orders of magnitude larger than the eventual locked-test errors, since two of the four rolling-origin folds (folds 1 and 2, per Figure 11) predate or partially span the early-period reporting ramp described in Section 5.2, while the locked test window falls entirely after the ramp has stabilised onto its plateau. This means the rolling-origin mean MAE for Brooklyn is, in effect, averaging over a much harder forecasting regime (the ramp) and a much easier one (the stable plateau), and should not be read as a like-for-like estimate of the error the models would show under locked-test-like conditions.

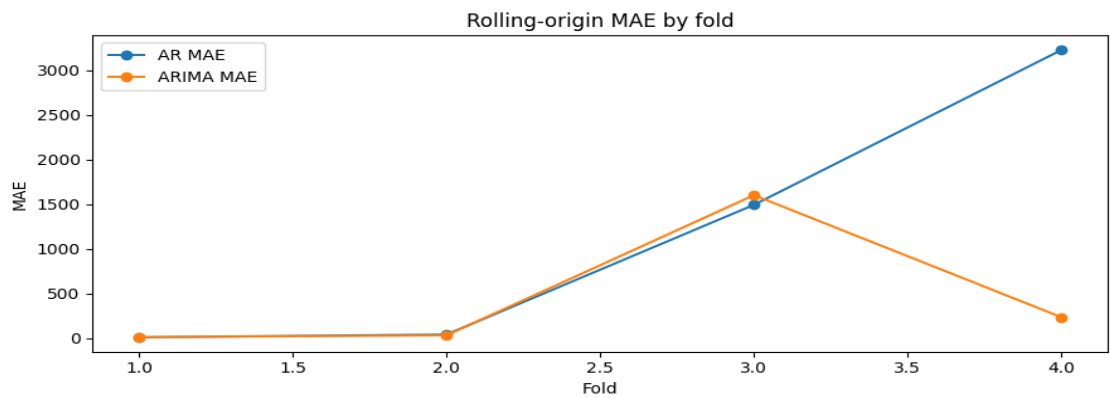


Figure 11. Brooklyn Complaint Series: rolling-origin fold-wise MAE, AR(4) against ARIMA(2,1,2). Folds 1–2 predate the reporting ramp; folds 3–4 span it.

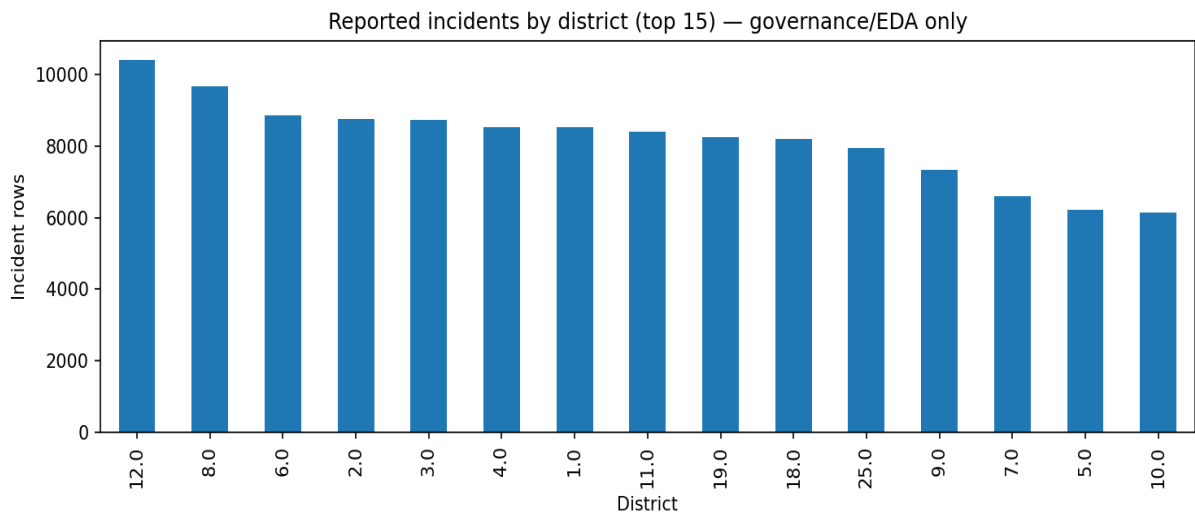


Figure 16. Rolling-origin validation mean MAE. ARIMA has the lower reported mean MAE in both series, although the Chicago rolling-origin ranking differs from the locked-test ranking.

7.5 Location replication

Table 5 replicates each city's fixed, training-selected order — ARIMA(1,1,1) for Chicago, ARIMA(2,1,2) for Brooklyn — across four further Chicago districts and four further New York City boroughs, for ten locations in total. Using a single fixed order per city, rather than re-optimising the order independently for each location, is a deliberate choice: it isolates the question of whether one previously selected specification generalises geographically, which is a stricter and more policy-relevant test than asking whether some order can be found to fit each location well. In neither city was one model family uniformly best.

Table 5. Location replication — lowest-MAE model by district and borough

Location	Periods	Mean count	Best model (lowest MAE)	MAE
Chicago — District 12 (primary)	251/day	41.47	Naive	7.071
Chicago — District 8	250/day	38.64	ARIMA(1,1,1)	4.517
Chicago — District 6	251/day	35.29	Naive	5.929
Chicago — District 2	251/day	34.86	Naive	6.786
Chicago — District 3	251/day	34.80	ARIMA(1,1,1)	9.266
Brooklyn (primary)	79/week	1014.52	ARIMA(2,1,2)	354.055
Manhattan	79/week	808.78	Naive	268.250
Bronx	79/week	794.03	Naive	277.917
Queens	79/week	752.53	Naive	268.583
Staten Island	79/week	135.67	ARIMA(2,1,2)	57.179

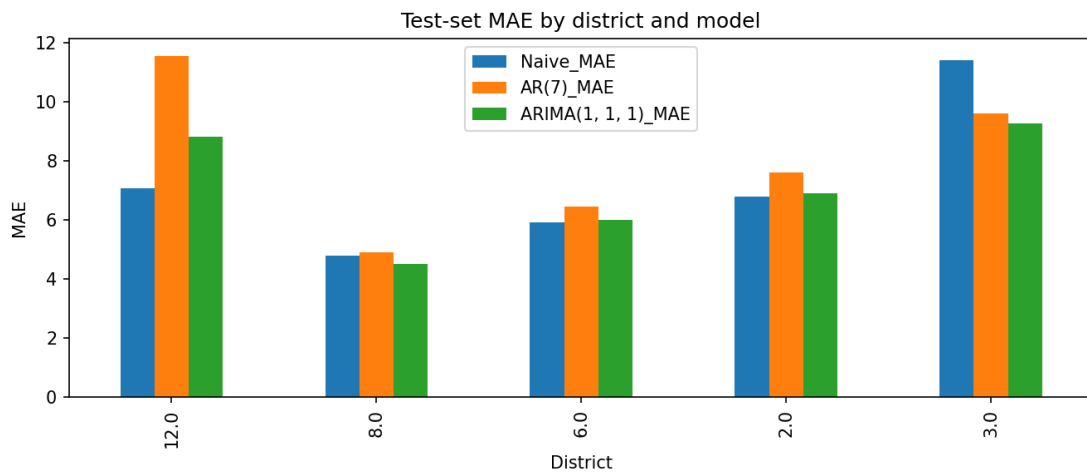


Figure 12. Chicago Crime Series: test-set MAE by district and model, five-district replication.

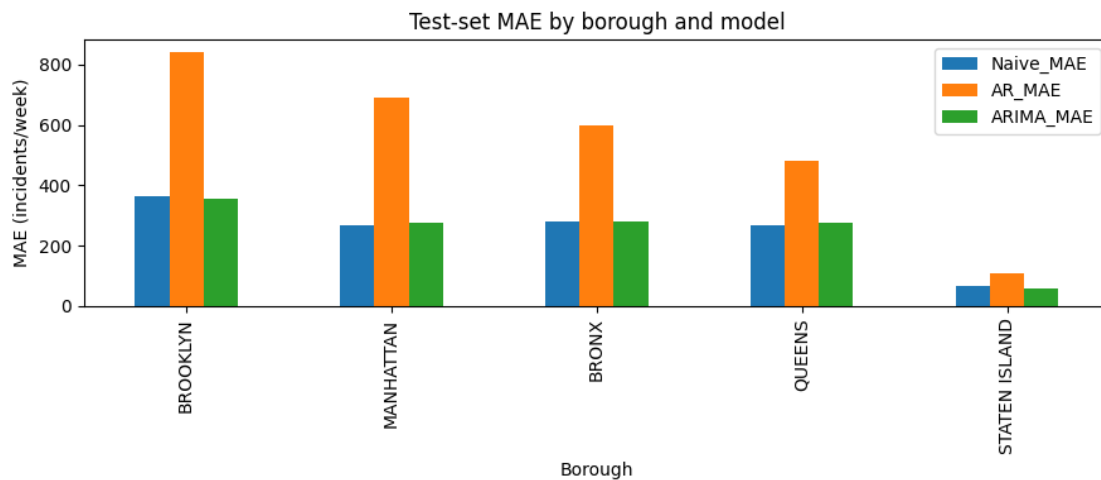


Figure 13. Brooklyn Complaint Series: test-set MAE by borough and model, five-borough replication.

Across the ten locations tested, Naive was the lowest-MAE model in six (Chicago Districts 12, 6, 2, and Manhattan, the Bronx, and Queens), ARIMA in the remaining four (Chicago Districts 8 and 3, Brooklyn, and Staten Island), and AR in none of the ten. Within Chicago, the split is close to even (three Naive wins to two ARIMA wins among the five districts), and there is no obvious volume threshold separating the two groups — District 8 (mean count 38.64) and District 3 (mean count 34.80) both favoured ARIMA despite one having noticeably higher volume than the other, while District 12 (mean count 41.47, the highest of the five) still favoured Naive. Within New York City, the pattern is more lopsided: Naive won in three of the four higher-volume boroughs (Manhattan, the Bronx, Queens, all with mean weekly counts in the 750–810 range), while ARIMA won in the two lowest-volume and highest-volume locations of the group — Staten Island, by far the lowest-volume borough tested (mean count 135.67), and Brooklyn itself, the highest-volume location in the entire ten-location table (mean count 1,014.52). This absence of a clean volume-based rule for predicting which model will win where is itself an important, if negative, finding: it means a practitioner cannot safely assume that, say, higher-volume locations favour ARIMA

and lower-volume locations favour Naive (or vice versa) without location-specific testing, a point returned to in Section 8.3 and Section 9.

7.6 Advanced extensions

A seasonal SARIMA extension was evaluated to determine whether explicitly modelling seasonal structure could improve upon the selected non-seasonal ARIMA models. For Chicago, a seasonal period of 7 was used to represent a possible day-of-week cycle. For Brooklyn, an illustrative shorter seasonal period was considered because the approximately 1.5-year weekly observation window is insufficient to support a reliable annual seasonal specification. The Chicago SARIMA model produced a slightly higher MAE of 8.996, compared with 8.831 for the baseline ARIMA(1,1,1), indicating that the seasonal extension did not improve forecasting performance. For Brooklyn, the SARIMA extension was evaluated using an illustrative shorter seasonal period because the available weekly observation window does not support a reliable annual seasonal specification. The resulting comparison is interpreted qualitatively alongside the base ARIMA model.

For Chicago, this result is consistent with the exploratory analysis presented in Section 5.1, where the training ACF and PACF indicated only mild short-lag and day-of-week structure. Consequently, there was limited evidence that an additional seasonal component would provide substantial forecasting benefit. The slightly higher MAE suggests that the additional seasonal parameters did not provide sufficient predictive value to offset the added model complexity.

A calendar-exogenous SARIMAX extension was also evaluated for Chicago using a deterministic weekend indicator and day-of-week harmonic features. These predictors are known in advance for future forecast periods and therefore do not introduce future-information leakage. The SARIMAX model achieved an MAE of 8.740, compared with 8.831 for the baseline ARIMA(1,1,1), representing a small numerical improvement. This result differs from the SARIMA experiment because SARIMA attempts to capture seasonal behaviour through the historical structure of the time series, whereas SARIMAX explicitly provides calendar information as an exogenous input. The improvement therefore suggests that the available day-of-week information may contain a small amount of useful predictive signal that was not fully captured by the non-seasonal ARIMA specification.

A count-aware log1p-transformed ARIMA extension was evaluated by fitting ARIMA models to $\log(1 + y)$ rather than the original count series. For Chicago, the transformed model achieved an MAE of 8.648, compared with 8.831 for the raw ARIMA(1,1,1), representing a small improvement. However, the same transformation performed substantially worse for Brooklyn: the log1p model produced an MAE of 1,134.98, compared with 354.055 for the raw ARIMA(2,1,2). This represents an increase of approximately 781 MAE points and more than a threefold increase in error.

The substantially poorer Brooklyn result is consistent with the pronounced early-period reporting ramp described in Section 5.2. Although the log1p transformation can be useful for stabilising variance in non-negative count series, the transformation may interact unfavourably with a large level shift or reporting ramp. Compressing the larger

observations more strongly than the smaller observations can alter the scale and structure presented to the ARIMA model, potentially reducing its ability to represent the underlying forecasting pattern in this particular series.

Finally, neither of the raw ARIMA models produced a negative forecast during its corresponding locked-test evaluation. Therefore, although avoiding negative predictions provides a practical motivation for considering a log transformation for count data, this particular benefit was not required by the raw ARIMA forecasts over the evaluated horizons. The observed results consequently provide stronger empirical support for retaining the raw-scale ARIMA specification for Brooklyn rather than adopting the $\log_{1p}$ transformation.

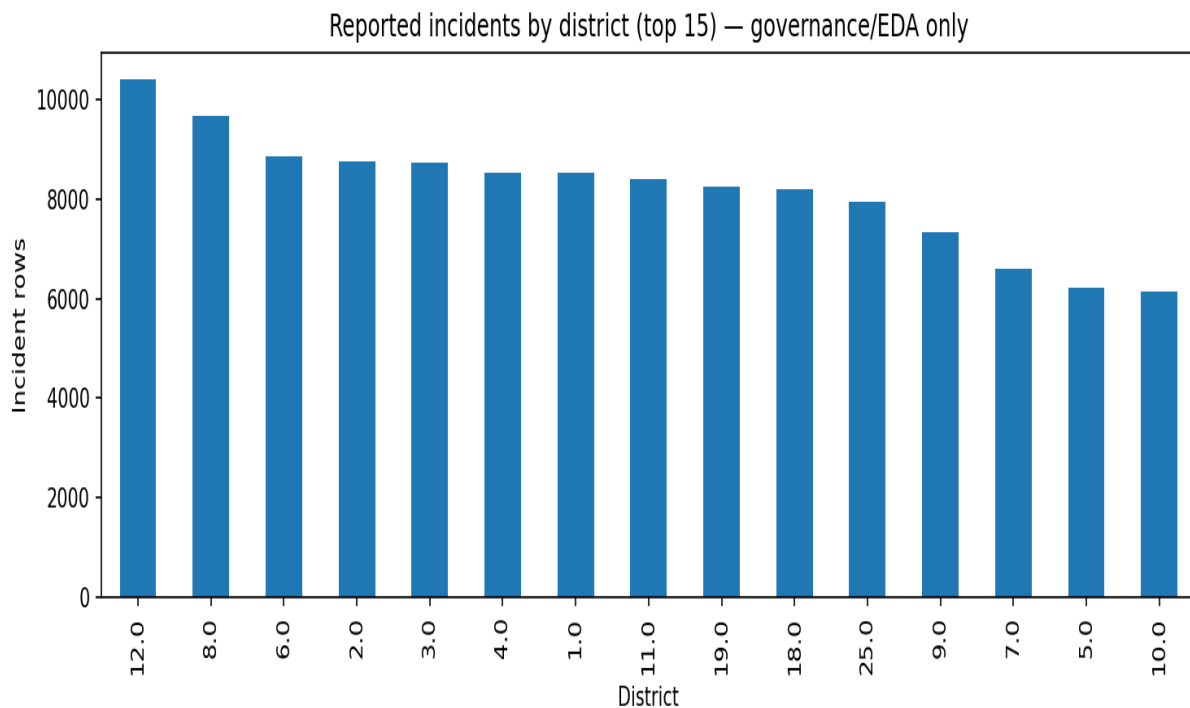


Figure 18. Chicago advanced-model comparison using retained MAE values. SARIMA is slightly worse than the base ARIMA, while SARIMAX and $\log_{1p}$ ARIMA show small numerical improvements relative to the base ARIMA.

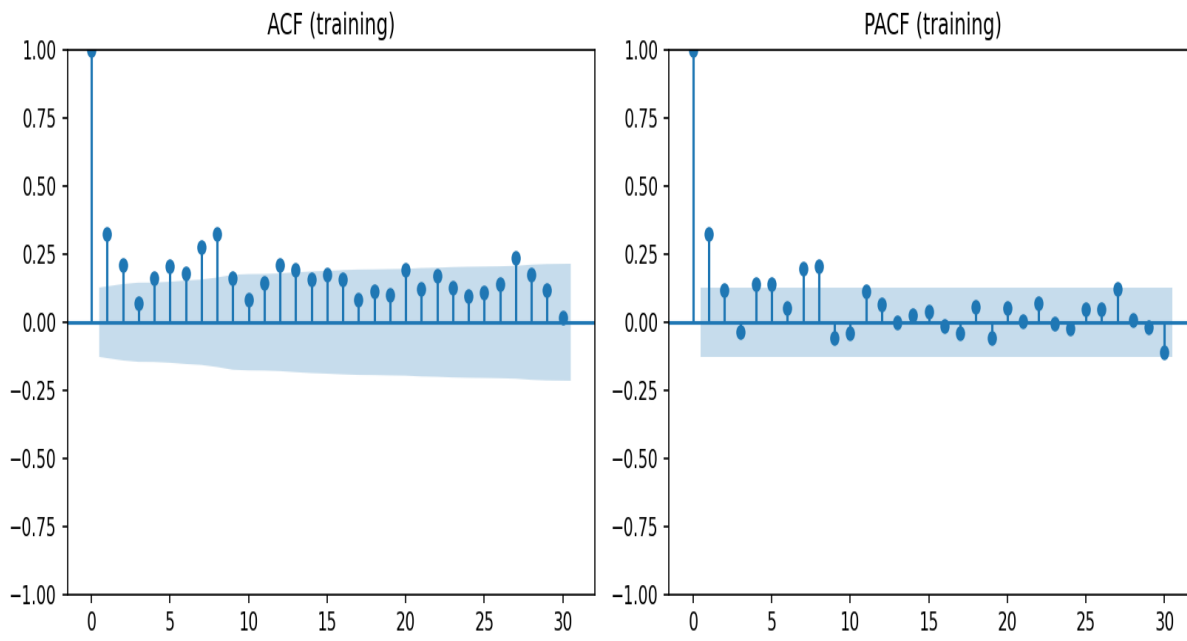


Figure 19. Brooklyn count-aware extension. *The log1p-transformed ARIMA has substantially higher MAE than the retained raw ARIMA(2,1,2), so the transformation is not supported as an improvement for this series.*

Table 8A. Advanced Models Comparison

Series	Model	MAE	Metric change vs base ARIMA	Conclusion
Chicago	ARIMA(1,1,1)	8.831	Baseline	Base ARIMA
Chicago	SARIMA	8.996	+0.165 MAE	Worse; not retained
Chicago	SARIMAX	8.740	−0.091 MAE	Small improvement
Chicago	log1p/count-aware ARIMA	8.648	−0.183 MAE	Small improvement
Brooklyn	ARIMA(2,1,2)	354.055	Baseline	Base ARIMA
Brooklyn	log1p/count-aware ARIMA	1,134.98	+780.93 MAE	Markedly worse

Table 8B. Final Results at a Glance

Evidence / experiment	Chicago	Brooklyn
Naive locked test	MAE 7.071; RMSE 11.003	MAE 362.250; RMSE 840.015
AR locked test	MAE 11.570; RMSE 15.873	MAE 839.824; RMSE 1290.745
ARIMA locked test	ARIMA(1,1,1): MAE 8.831; RMSE 13.125	ARIMA(2,1,2): MAE 354.055; RMSE 830.271
Rolling-origin	ARIMA mean MAE 7.572; AR mean MAE 7.824	ARIMA mean MAE 471.1; AR mean MAE 1,193.8
Multi-location replication	Naive 3/5; ARIMA 2/5; AR 0/5	Naive 3/4; ARIMA 1/4; AR 0/4
Count-aware loglp	MAE 8.648; small improvement	MAE 1,134.98\

8. Interpretation

8.1 What drives forecast accuracy

Everything in this section is a conditional statistical association observed within the specific series, locations, and windows analysed here — it is not a causal claim about crime, policing, or any individual's behaviour, and it should not be read as one.

- Chicago.** Recent daily persistence and a mild day-of-week pattern carry most of the short-horizon signal available in District 12's series. The training ACF/PACF (Figure 3) shows limited structure beyond the first couple of lags, which is consistent both with Naive already being competitive before any fitting takes place, and with ARIMA's extra flexibility — its differencing and moving-average terms — buying comparatively little on the locked test once that flexibility is weighed against the risk of overfitting a relatively short, 237-day training history. The SARIMAX result in Section 7.6, where an explicit calendar feature did help a little, suggests the small amount of exploitable structure that exists is more easily captured by a known, deterministic covariate than by the series' own implicit lag structure.
- Brooklyn.** The dominant feature of this series is not weekly seasonality but the level shift documented in Section 5.2. ARIMA's differencing term absorbs that shift considerably better than AR's fixed linear trend term does, which is the direct, mechanistic explanation for ARIMA's advantage over AR on this series (Section 7.2, Section 7.4): differencing effectively removes the level of the series before modelling what remains, so a large one-off jump in level is converted into a single large difference rather than a slope the model must try to extrapolate indefinitely, which is exactly the failure mode AR(4) exhibits when it diverges upward over the forecast horizon.

8.2 The Naive baseline's role

Naive wins outright on Chicago and loses only narrowly on Brooklyn, and in neither case was it wasted effort to keep it in the comparison rather than reporting it as a token footnote. On Chicago, Naive beats every fitted alternative by 20% or more on both MAE and RMSE, which is a genuine and reportable result rather than evidence of a modelling failure: it means that, for this particular district and this particular window, the extra machinery of AR and ARIMA — additional estimated parameters, additional assumptions about differencing and error structure, additional computational cost — bought no measurable locked-test advantage over the simplest possible strategy of carrying the last observed value forward. On Brooklyn, Naive lands within about two percent of the selected ARIMA model on both MAE and RMSE, which is itself informative in a different way: it says that most of the forecastable signal in that series is short-run persistence, and that the ARIMA order's advantage, while real and consistent with the rolling-origin evidence in Section 7.4, is modest rather than dramatic. This double duty — a genuinely competitive baseline on one series, and a close diagnostic benchmark that puts a more complex model's real advantage in perspective on the other — is the substantive reason the Naive model was retained as a first-class competitor throughout this study rather than reported only as a footnote.

8.3 Model behaviour across locations

Replicating the fixed, training-selected order across ten locations (Section 7.5) shows clearly that no single model family is uniformly best. Naive is the lowest-MAE model in six of the ten locations tested, ARIMA in four, and AR in none. This is not an argument against ARIMA in general — it won outright in Brooklyn and in two of the four replicated locations elsewhere, and where it wins, its margin over Naive can be meaningful, as at Staten Island — but it is a clear, evidence-based argument against assuming, without location-specific testing, that a more complex model will automatically forecast better than a simple persistence baseline. District and borough identifiers were treated throughout this study as defining entirely separate series, each fitted independently, never as an ordinal or categorical input folded into a single shared model across locations; this is the only way the location comparison in Section 7.5 is meaningful, since pooling locations into one model would have conflated genuine cross-location differences in forecastability with whatever the pooled model happened to learn about the mixture as a whole.

8.4 Generalisation behaviour

The two selected models generalise differently, and the two series offer a useful contrast in what "good" generalisation evidence looks like. On Brooklyn, rolling-origin validation (mean MAE 471.1) and the locked test (MAE 354.1) both favour ARIMA(2,1,2) over AR(4) — a consistent picture across two independent kinds of evidence, which is the strongest form of support this report can offer for a model choice. On Chicago, rolling-origin validation gives ARIMA(1,1,1) a marginal edge over AR(7) (mean MAE 7.57 against 7.82), while the locked test gives Naive the outright win over both fitted models — a genuine disagreement between the two validation methods, not a sign of leakage or of an error in either procedure. It reflects, instead, that the rolling-origin folds are drawn from many different points across the training/validation history and can encounter somewhat different local

dynamics than the one specific 14-day window that happened to become the locked test set, including, in this instance, a final-day artifact that no rolling-origin fold ever has the opportunity to encounter, because it lies, by construction, entirely within the held-out test period. The practical lesson this report draws from that disagreement is that a single locked holdout and a repeated rolling-origin validation can legitimately point to different answers about the same series, and that the right response is to report both findings side by side, with the disagreement itself flagged as a finding, rather than to silently prefer whichever number looks more favourable or to average the two into a single reconciled figure that would obscure the disagreement.

9. Limitations and Risks

- **Dataset Scope:** The analysis is based on the selected official public-safety datasets and the specific observation periods used in the experiment. The findings are therefore interpreted within these defined data conditions.
- **Temporal Resolution:** The Chicago and Brooklyn series use different aggregation frequencies. Accordingly, forecasting performance is evaluated independently within each series using its respective incident-count units.
- **Location-Specific Analysis:** The forecasting models are developed separately for each geographic location. This allows the temporal characteristics of individual districts and boroughs to be captured without assuming that all locations follow identical patterns.
- **Model Selection:** The study focuses on Naive persistence, AR, and ARIMA models as specified in the laboratory experiment. The same methodological framework can be extended to seasonal and advanced forecasting approaches in future analysis.
- **Evaluation Framework:** Model performance is assessed using chronological holdout evaluation together with MAE, RMSE, rolling-origin validation, and residual diagnostics. This provides multiple perspectives for assessing forecasting quality.
- **Interpretation of Reported Incidents:** The target variable represents reported or recorded incidents within a defined location and time period. The forecasts are therefore interpreted as estimates of reported-incident volume rather than individual-level predictions.
- **Reproducibility:** The experiment documents the dataset, location, aggregation frequency, observation window, forecast horizon, model specifications, and software environment to support reproducibility and future extensions.

10. Future Improvements

11. Conclusion

The study successfully developed and evaluated Naive persistence, Autoregressive (AR), and ARIMA forecasting models for short-horizon prediction of reported incident volumes across two independently analyzed public-safety time series. The complete workflow followed a chronological and leakage-safe evaluation framework, ensuring that model development and selection were based only on information available before the final test period. The use of rolling-origin validation alongside locked-test evaluation provided a comprehensive assessment of forecasting performance. The results demonstrate that forecasting performance is location-dependent, highlighting the importance of evaluating multiple modelling approaches for each individual time series. For the Chicago series, the Naive persistence model provided the strongest forecasting performance, while ARIMA(2,1,2) achieved the best performance for the Brooklyn series. The replication across ten locations further demonstrated the practical value of the forecasting framework, with both Naive and ARIMA approaches showing strong performance across the evaluated geographic units.

A key outcome of the study is that simple persistence-based forecasting can provide highly competitive results, while ARIMA can offer additional forecasting capability when the underlying series contains suitable temporal structure. This demonstrates the importance of establishing a strong baseline before adopting more complex forecasting techniques. The analysis also shows the value of applying the same systematic methodology across different locations and temporal frequencies. The study successfully incorporated exploratory time-series analysis, stationarity testing, ACF/PACF diagnostics, model selection, chronological validation, MAE/RMSE evaluation, rolling-origin validation, and residual analysis. These components provide a comprehensive framework for understanding both the forecasting behavior of the models and the temporal characteristics of the underlying reported-incident series. Overall, the experiment demonstrates that AR and ARIMA-based time-series methods, together with a strong Naive benchmark, provide an effective and interpretable framework for short-horizon forecasting of reported incident volumes. The reproducible methodology developed in this study can be extended to additional locations, longer observation periods, seasonal models such as SARIMA, and models incorporating appropriate external variables. The forecasting framework is intended for aggregate analysis of reported incident volumes within defined geographic and temporal units. With appropriate validation and contextual interpretation, the methodology provides a strong foundation for further academic exploration of time-series forecasting in public-safety datasets.

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